WORKED SOLUTION

STEP I 2010, Question 10

Mechanics • Time-displacement

STEP I 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q10

The key idea

Everything follows from differentiating $\mathbf{r}$ and taking dot products: the angle between two vectors comes from $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$, and perpendicularity means a zero dot product. For the second particle, the useful observation is that Q at time t is exactly where P was at time $t-T$. Writing the position as a complex number, $z = \mathrm{e}^{(1+\mathrm{i})t}$, turns the last part into a one-line calculation.

Solution

Velocity and acceleration

Write $\mathbf{r} = \mathrm{e}^t(\cos t,\ \sin t)$. By the product rule,

$$\begin{aligned}\mathbf{v} = \dot{\mathbf{r}} &= \mathrm{e}^t(\cos t - \sin t,\ \sin t + \cos t),\\ \mathbf{a} = \dot{\mathbf{v}} &= \mathrm{e}^t((\cos t - \sin t) + (-\sin t - \cos t),\ (\sin t + \cos t) + (\cos t - \sin t))\\ &= \mathrm{e}^t(-2\sin t,\ 2\cos t).\end{aligned}$$

The angle between velocity and displacement

The magnitudes are $|\mathbf{r}| = \mathrm{e}^t$ and

$$|\mathbf{v}|^2 = \mathrm{e}^{2t}[(\cos t - \sin t)^2 + (\sin t + \cos t)^2] = \mathrm{e}^{2t}\cdot 2,$$

since the cross terms $\mp 2\sin t\cos t$ cancel. So $|\mathbf{v}| = \sqrt{2}\,\mathrm{e}^t$. The dot product is

$$\mathbf{r}\cdot\mathbf{v} = \mathrm{e}^{2t}[\cos t(\cos t - \sin t) + \sin t(\sin t + \cos t)] = \mathrm{e}^{2t}(\cos^2 t + \sin^2 t) = \mathrm{e}^{2t}.$$

Hence, if θ is the angle between $\mathbf{r}$ and $\mathbf{v}$,

$$\cos\theta = \frac{\mathbf{r}\cdot\mathbf{v}}{|\mathbf{r}||\mathbf{v}|} = \frac{\mathrm{e}^{2t}}{\mathrm{e}^t\cdot\sqrt{2}\,\mathrm{e}^t} = \frac{1}{\sqrt{2}},$$

so $\theta = \frac{\pi}{4}$ for every t, as required. (Neither vector is ever zero, since $\mathrm{e}^t > 0$, so the angle is always defined.)

The acceleration is perpendicular to the displacement

$$\mathbf{r}\cdot\mathbf{a} = e^{2t}[-2\sin t\cos t + 2\sin t\cos t] = 0,$$

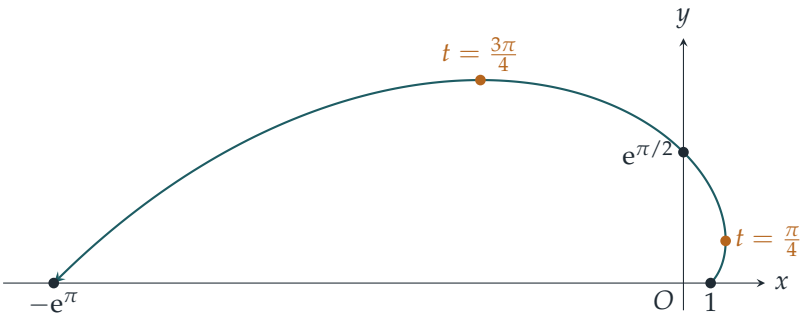
and both vectors are non-zero, so $\mathbf{a}$ is always perpendicular to $\mathbf{r}$. Notice also that $\mathbf{a} = 2e^t(-\sin t, \cos t)$ is $\mathbf{r}$ rotated by $\frac{\pi}{2}$ and doubled.

The path for $0 \le t \le \pi$

In polar coordinates the path is $r = e^\theta$ with $\theta = t$: the distance from O grows exponentially as the particle turns anticlockwise, an *equiangular* (logarithmic) spiral. Useful landmarks:

- $t = 0$: the point $(1, 0)$, moving in direction $(1, 1)$, at $\frac{\pi}{4}$ to the x-axis.
- $t = \frac{\pi}{4}$: $\dot{x} = e^t(\cos t - \sin t) = 0$, so x is greatest, at the point $\left(\frac{1}{\sqrt{2}}e^{\pi/4}, \frac{1}{\sqrt{2}}e^{\pi/4}\right) \approx (1.55, 1.55)$.
- $t = \frac{\pi}{2}$: the point $(0, e^{\pi/2}) \approx (0, 4.81)$.
- $t = \frac{3\pi}{4}$: $\dot{y} = e^t(\sin t + \cos t) = 0$, so y is greatest, at $\left(-\frac{1}{\sqrt{2}}e^{3\pi/4}, \frac{1}{\sqrt{2}}e^{3\pi/4}\right) \approx (-7.46, 7.46)$.
- $t = \pi$: the point $(-e^\pi, 0) \approx (-23.1, 0)$, crossing the axis at $\frac{\pi}{4}$, heading down and to the left.

At every point the path crosses the radius from O at $\frac{\pi}{4}$, which fixes its shape.



The second particle

? **Ask yourself** Where is Q at time t, in terms of P's motion?

Q reaches each point T later than P, so at time t it is where P was at time $t - T$:

$$\mathbf{r}_Q(t) = \mathbf{r}(t-T) = e^{t-T}\big(\cos(t-T),\ \sin(t-T)\big).$$

Then

$$\begin{aligned}|PQ|^2 &= |\mathbf{r}(t)|^2 + |\mathbf{r}(t-T)|^2 - 2\,\mathbf{r}(t)\cdot\mathbf{r}(t-T)\\ &= \mathrm{e}^{2t} + \mathrm{e}^{2t-2T} - 2\mathrm{e}^{2t-T}\big(\cos t\cos(t-T) + \sin t\sin(t-T)\big)\\ &= \mathrm{e}^{2t}\big(1 - 2\mathrm{e}^{-T}\cos T + \mathrm{e}^{-2T}\big),\end{aligned}$$

using $\cos t\cos(t-T) + \sin t\sin(t-T) = \cos\big(t-(t-T)\big) = \cos T$. Therefore

$$|PQ| = k\,\mathrm{e}^{t}, \qquad k = \sqrt{1 - 2\mathrm{e}^{-T}\cos T + \mathrm{e}^{-2T}},$$

and k depends only on T, not on t. So the distance between P and Q is proportional to e^t, as required. (The bracket is $|1-\mathrm{e}^{-T}\mathrm{e}^{\mathrm{i}T}|^2$, which is positive for $T \neq 0$, since $\mathrm{e}^{-T} \neq 1$; so $k > 0$.)

Complex-number version. With $z = x + \mathrm{i}y$, P is at $z_P = \mathrm{e}^t\mathrm{e}^{\mathrm{i}t} = \mathrm{e}^{(1+\mathrm{i})t}$ and Q at $z_Q = \mathrm{e}^{(1+\mathrm{i})(t-T)}$. So

$$z_P - z_Q = \mathrm{e}^{(1+\mathrm{i})t}\big(1 - \mathrm{e}^{-(1+\mathrm{i})T}\big), \qquad |z_P - z_Q| = \mathrm{e}^t\,\big|1 - \mathrm{e}^{-(1+\mathrm{i})T}\big|.$$

This also shows that $\overrightarrow{QP}$ is a fixed complex multiple of z_P, so the segment PQ makes a constant angle with OP: the triangle OQP keeps the same shape and just grows and turns.

Checks

- Differentiating $z = \mathrm{e}^{(1+\mathrm{i})t}$ gives $\dot z = (1+\mathrm{i})z$ and $\ddot z = (1+\mathrm{i})^2 z = 2\mathrm{i}z$. Multiplying by $1+\mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$ rotates by $\frac{\pi}{4}$, and multiplying by $2\mathrm{i}$ rotates by $\frac{\pi}{2}$: both of the first results at a glance.
- Numerically, for random t the angle between $\mathbf{r}$ and $\mathbf{v}$ came out as $0.785398\ldots = \frac{\pi}{4}$ and $\mathbf{r}\cdot\mathbf{a} = 0$. With $T = 1$, $|PQ|/\mathrm{e}^t = 0.85895\ldots$ at every t tested, matching $\sqrt{1 - 2\mathrm{e}^{-1}\cos 1 + \mathrm{e}^{-2}}$.
- As $T \to 0$, $k \approx T\sqrt{2}$, so $|PQ| \approx \sqrt{2}\,\mathrm{e}^t T = |\mathbf{v}|T$, which is the distance travelled in time T, as it should be.

Common slips

- Taking Q's position as $\mathbf{r}(t+T)$: Q is *behind* P, so it is at $\mathbf{r}(t-T)$. (The answer is still proportional to e^t, but with a different constant.)
- Sketching a spiral that turns clockwise, or showing the path through the origin. The distance from O is $\mathrm{e}^t \geq 1$ and increases as the particle moves anticlockwise.

Beyond STEP

The curve $r = \mathrm{e}^\theta$ is the logarithmic spiral: every scaling of it about O is a rotation of it, which is why the crossing angle is constant and why OQP keeps its shape. In first-year

courses, motion in a plane is often handled with polar unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}$, in which the radial acceleration $\ddot{r} - r\dot{\theta}^2$ vanishes here, giving the perpendicularity directly. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops polar coordinates for exactly this kind of problem.

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